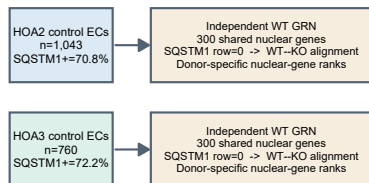


A

Donor-separated in silico perturbation design

HOA2 and HOA3 networks were fitted and perturbed independently

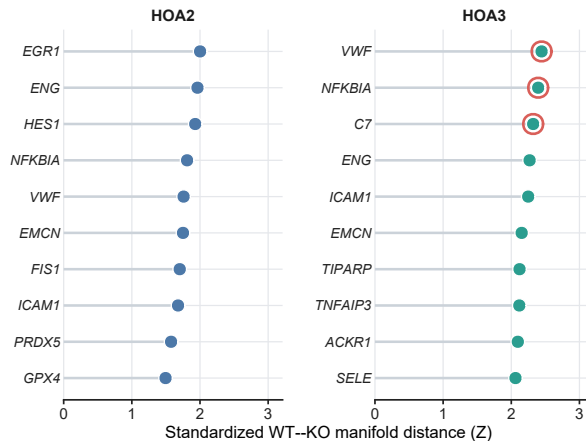


Shared nuclear feature universe: donor matrices and networks were not pooled
HOA1 ineligible (26 ECs) | two parameter profiles | 20 matched-gene perturbations | no experimental KO

B

Predicted nuclear differential regulation

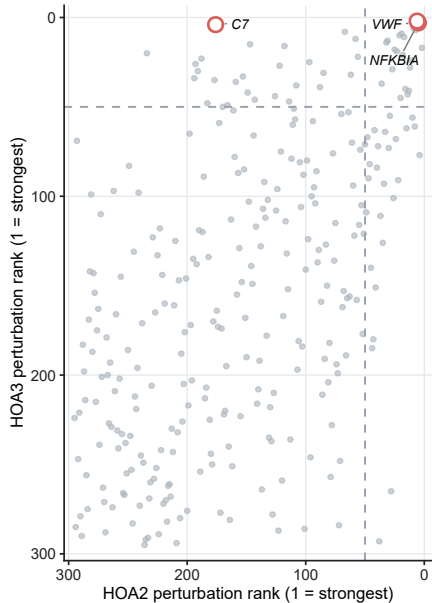
Top 10 downstream ranks per donor; red rings denote donor-specific FDR<0.05



C

Cross-donor rank reproducibility

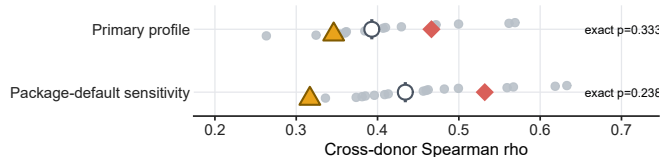
Spearman rho=0.555; replicated FDR hits=0
Red rings: HOA3-only hits



D

Receptor-context specificity calibration

Grey: matched genes; gold triangle: CALCOCO2; open circle: median; red diamond: SQSTM1



E

Matched-null pathway rank recovery

Red outline: empirical FDR<0.05 in one donor only; no pathway replicated

